Degree and Community Information in Random-Walk Network Embeddings

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Abstract

We study how node degree and community membership enter population network embeddings under the degree-corrected stochastic block model. The normalized-adjacency eigenspace and the stationary random-walk joint probability matrix both have an exact radial-angular structure: node propensity controls radius within each community, while normalized direction depends only on community. Skip-gram with negative sampling changes this structure. At unigram exponent one, marginal normalization cancels node-specific degree from the population score. At other exponents, the optimal score contains an additive context-degree term rather than a radial multiplier. Individual embedding norms and angles are therefore identifiable only after tying or balancing the two factors. These results separate the geometry of raw random-walk pairs from the geometry of the score matrix optimized by standard embedding algorithms.

1 Introduction

Spectral embeddings under the degree-corrected stochastic block model (DCSBM) have a characteristic ray geometry. Rows associated with nodes in one community point in a common direction, while their lengths vary with degree propensity [Qin and Rohe, 2013]. Row normalization removes this within-community variation and permits community recovery.

Random-walk embedding methods combine a walk-based sampling distribution with skip-gram training. The matrix-factorization view of skip-gram identifies a shifted pointwise mutual information score [Levy and Goldberg, 2014, Qiu et al., 2018]. Recent analyses establish community-recovery guarantees and spectral connections for DeepWalk and node2vec under stochastic block models [Zhang and Tang, 2022, Kojaku et al., 2024, Davison et al., 2024]. The location and identifiability of degree information within these embeddings remain less clear.

We derive population formulas that distinguish three operators. The normalized-adjacency eigenspace has row norm proportional to the square root of expected degree within each community. The stationary joint probability of random-walk pairs has row norm proportional to expected degree in every minimal exact factorization. The unconstrained skip-gram score cancels node-specific degree when the negative-sampling exponent is one and introduces a one-sided additive term for other exponents.

Our random-walk calculations apply to DeepWalk, equivalently node2vec with return and in–out parameters $p = q = 1$. General node2vec walks are second-order Markov chains on directed edges and require a separate lumpability argument.

2 DCSBM setup

Let n nodes belong to K communities. Write $g_i \in \{1, \ldots, K\}$ for the community of node i, and let $Z \in \{0, 1\}^{n \times K}$ be the membership matrix. Each node has a degree propensity $\theta_i > 0$, collected in $\Theta = \operatorname{diag}(\theta_1, \ldots, \theta_n)$. Let $B \in \mathbb{R}^{K \times K}$ be symmetric with nonnegative entries. We assume that every community is nonempty, that the block degree factors defined below are positive, and that $\theta_i \theta_j B_{g_i g_j} \leq 1$.

The rank-K population proxy is

$$\Omega = \Theta Z B Z^\top \Theta, \qquad \Omega_{ij} = \theta_i \theta_j B_{g_i g_j}. \tag{1}$$

This is the standard multiplicative DCSBM structure [Karrer and Newman, 2011].

Define the block propensity masses and block degree factors by

$$s_a = \sum_{i: g_i = a} \theta_i, \qquad \kappa_a = (Bs)_a = \sum_{b=1}^{K} B_{ab} s_b. \tag{2}$$

The population expected degree is

$$\delta_i = (\Omega \mathbf{1})_i = \theta_i \kappa_{g_i}. \tag{3}$$

Let $\mathcal{D} = \operatorname{diag}(\delta_1, \ldots, \delta_n)$ and

$$\mathrm{vol} = \sum_i \delta_i = s^\top B s. \tag{4}$$

Remark 1 (Diagonal convention)**.** *Equation* (1) *includes the diagonal of the low-rank population proxy. A loopless DCSBM has mean adjacency* $\Omega - \operatorname{diag}(\Omega)$*. If we retain the proxy normalization* $\mathcal{D}$*, removing the diagonal changes the normalized operator by a diagonal matrix with operator norm*

$$\max_i \frac{\theta_i B_{g_i g_i}}{\kappa_{g_i}}. \tag{5}$$

The relative difference between the proxy degree and the loopless expected degree is bounded by the same quantity. Both differences vanish when the propensities remain bounded and the block degree masses grow.

3 Multiplicative block matrices

Lemma 2 (Multiplicative block matrices produce community rays)**.** *Let*

$$M = \operatorname{diag}(r) Z H Z^\top \operatorname{diag}(t), \tag{6}$$

where $r_i, t_i > 0$*,* $H \in \mathbb{R}^{K \times K}$*, and* $h = \operatorname{rank}(M)$*. Suppose that* $M = UV^\top$ *is a minimal exact factorization with* $\operatorname{rank}(U) = \operatorname{rank}(V) = h$*. Then matrices* $C, E \in \mathbb{R}^{K \times h}$ *exist such that*

$$u_i = r_i c_{g_i}, \qquad v_i = t_i e_{g_i}, \tag{7}$$

where lowercase letters denote rows of their corresponding matrices. Consequently,

$$\|u_i\| = r_i \|c_{g_i}\|, \qquad \frac{u_i}{\|u_i\|} = \frac{c_{g_i}}{\|c_{g_i}\|}, \tag{8}$$

and

$$\cos(u_i, u_j) = \frac{c_{g_i}^\top c_{g_j}}{\|c_{g_i}\| \|c_{g_j}\|}. \tag{9}$$

Proof. The column space of M is contained in the column space of $\mathrm{diag}(r)Z$. Minimality gives $\mathrm{col}(U) = \mathrm{col}(M)$, so $U = \mathrm{diag}(r)ZC$ for some C. Applying the same argument to $M^\top$ gives $V = \mathrm{diag}(t)ZE$. Equations (7)–(9) follow row by row. $\square$

The lemma does not require M to be positive semidefinite or its factorization to be unique. An invertible change of factorization changes the community vectors but preserves the multiplier r_i within each community. Two communities have distinct normalized directions when their community vectors are nonzero and are not positive scalar multiples of one another. Minimality, or an equivalent balancing condition, rules out canceling components in unused dimensions.

4 Normalized spectral embeddings

Define the population normalized adjacency

$$\mathcal{S} = \mathcal{D}^{-1/2}\Omega\mathcal{D}^{-1/2}. \tag{10}$$

Let

$$Q = \Theta^{1/2} Z \,\mathrm{diag}(s)^{-1/2}. \tag{11}$$

Its columns are orthonormal because $Q^\top Q = I_K$. Define the block operator

$$H = \mathrm{diag}(s)^{1/2}\,\mathrm{diag}(\kappa)^{-1/2} B \,\mathrm{diag}(\kappa)^{-1/2}\,\mathrm{diag}(s)^{1/2}. \tag{12}$$

Direct multiplication gives

$$\mathcal{S} = QHQ^\top. \tag{13}$$

Theorem 3 (Degree controls radius in the population eigenspace)**.** *Suppose that B has rank K. Let $H = V\Lambda V^\top$, with V orthogonal. The K nonzero eigenvectors of $\mathcal{S}$ can be collected as $U = QV$. For node i in community $a = g_i$,*

$$u_i = \sqrt{\frac{\theta_i}{s_a}}\, v_a, \tag{14}$$

where v_a is row a of V. Therefore,

$$\|u_i\|^2 = \frac{\theta_i}{s_a} = \frac{\delta_i}{s_a\kappa_a}, \qquad \frac{u_i}{\|u_i\|} = v_a. \tag{15}$$

Proof. Equation (13) and $Q^\top Q = I_K$ imply

$$\mathcal{S}(QV) = QHV = QV\Lambda. \tag{16}$$

Equation (14) follows from the definition of Q. The norm formula uses $\|v_a\| = 1$. $\square$

Using all K population eigenvectors, nodes in one community have cosine one, while nodes in different communities have cosine zero because the rows of V are orthonormal. If an embedding drops the trivial eigenvector or retains fewer dimensions, cross-community angles need not be right angles, but they remain functions of the two community labels.

Corollary 4 (Within-community degree recovery)**.** *Within each community,*

$$\delta_i \propto \|u_i\|^2. \tag{17}$$

Across communities, recovery of δ_i requires the block calibration factor $s_a\kappa_a$.

Many matrix-factorization embeddings weight eigenvectors by eigenvalues. Define

$$x_i = e_i^\top U \,|\Lambda|^{1/2}\,, \tag{18}$$

where $|\Lambda|$ applies absolute value to each eigenvalue. Then

$$x_i = \sqrt{\frac{\theta_i}{s_a}}\, v_a \,|\Lambda|^{1/2}\,. \tag{19}$$

Direction remains community-specific, and norm remains proportional to $\sqrt{\theta_i}$ within a community. If $H \succeq 0$, then

$$\cos(x_i, x_j) = \frac{B_{ab}}{\sqrt{B_{aa}B_{bb}}}, \qquad a = g_i, \quad b = g_j. \tag{20}$$

For indefinite H, the same calculation replaces H by $V\,|\Lambda|\,V^\top$, and the angle remains block-specific.

5 Raw random-walk pairs

Let

$$T = \mathcal{D}^{-1}\Omega \tag{21}$$

be the population simple-random-walk transition matrix. For $a = g_i$ and $b = g_j$,

$$T_{ij} = \frac{\theta_j B_{ab}}{\kappa_a}. \tag{22}$$

The induced block transition matrix is

$$W_{ab} = \sum_{j:g_j=b} T_{ij} = \frac{B_{ab}s_b}{\kappa_a}. \tag{23}$$

Lemma 5 (Endpoint distribution at each lag)**.** *For every integer $r \geq 1$,*

$$(T^r)_{ij} = \frac{\theta_j}{s_b}(W^r)_{ab}. \tag{24}$$

Proof. Equation (24) holds for $r = 1$ by Equation (23). Assume that it holds at lag r. Grouping the intermediate node by community in $(T^{r+1})_{ij} = \sum_\ell (T^r)_{i\ell}T_{\ell j}$ gives the formula at lag $r+1$. □

Let $a_1, \ldots, a_L \geq 0$ be context-window weights with $\sum_r a_r = 1$, and define

$$F = \sum_{r=1}^{L} a_r W^r. \tag{25}$$

The stationary distribution is

$$\pi_i = \frac{\delta_i}{\mathrm{vol}}. \tag{26}$$

Define the stationary joint probability of the ordered positive pair (i, j) by

$$J_{ij} = \pi_i \sum_{r=1}^{L} a_r (T^r)_{ij}. \tag{27}$$

Table 1: Population radial exponents for three embedding operators.

Population object	Row form within block a	Within-block prediction
Normalized-adjacency eigenspace	$u_i = \sqrt{\theta_i}\, c_a$	$\|u_i\| \propto \sqrt{\delta_i}$
Raw adjacency or stationary pairs	$u_i = \theta_i c_a$	$\|u_i\| \propto \delta_i$
SGNS score with $\alpha = 1$	$u_i = c_a$	no degree signal

Theorem 6 (Raw stationary pairs have radial-angular form). *For $a = g_i$ and $b = g_j$,*

$$J_{ij} = \theta_i \theta_j G_{ab}, \qquad G_{ab} = \frac{\kappa_a}{\text{vol } s_b} F_{ab}. \tag{28}$$

The random walk is reversible, so G and J are symmetric. In matrix form,

$$J = \Theta Z G Z^\top \Theta. \tag{29}$$

Every minimal exact factorization $J = UV^\top$ therefore has

$$u_i = \theta_i c_{g_i}, \qquad v_i = \theta_i e_{g_i}. \tag{30}$$

Proof. Substituting Equations (3), (24), and (25) into Equation (27) yields

$$J_{ij} = \frac{\theta_i \kappa_a}{\text{vol}} \frac{\theta_j}{s_b} F_{ab}. \tag{31}$$

Reversibility gives $J_{ij} = J_{ji}$. Lemma 2 gives Equation (30). □

Within community a,

$$\|u_i\| = \theta_i \|c_a\| = \frac{\delta_i}{\kappa_a} \|c_a\|. \tag{32}$$

The raw stationary pair matrix therefore gives the strongest exact version of degree in radius and community in direction. It describes pair probabilities before marginal normalization and the entrywise logarithm.

6 Skip-gram with negative sampling

Let $p^+_{ij} = J_{ij}$ be the positive-pair probability. Suppose skip-gram with negative sampling (SGNS) draws k negative contexts per center from

$$q_j = \frac{\delta_j^\alpha}{Z_\alpha}, \qquad Z_\alpha = \sum_\ell \delta_\ell^\alpha. \tag{33}$$

The expected negative weight for ordered pair (i, j) is

$$p^-_{ij} = k \pi_i q_j. \tag{34}$$

Theorem 7 (Unconstrained SGNS score). *For positive weights p^+_{ij} and p^-_{ij}, the pairwise SGNS objective*

$$\ell_{ij}(x) = p^+_{ij} \log \sigma(x) + p^-_{ij} \log \sigma(-x) \tag{35}$$

has the unique maximizer

$$x^\star_{ij} = \log \frac{p^+_{ij}}{p^-_{ij}}. \tag{36}$$

For the DCSBM walk and every block pair with $F_{ab} > 0$,

$$x^\star_{ij} = H^{(\alpha)}_{ab} + (1-\alpha)\log\theta_j, \tag{37}$$

where

$$H^{(\alpha)}_{ab} = \log F_{ab} - \log s_b - \alpha\log\kappa_b + \log Z_\alpha - \log k. \tag{38}$$

Proof. Differentiation gives

$$\ell'_{ij}(x) = \frac{p^+_{ij} - p^-_{ij}e^x}{1+e^x} \tag{39}$$

and

$$\ell''_{ij}(x) = -\frac{(p^+_{ij} + p^-_{ij})e^x}{(1+e^x)^2} < 0. \tag{40}$$

Thus $e^{x^\star} = p^+_{ij}/p^-_{ij}$. Substituting

$$\sum_r a_r (T^r)_{ij} = \frac{\theta_j}{s_b}F_{ab}, \qquad \delta_j = \theta_j\kappa_b \tag{41}$$

gives Equations (37) and (38). □

Corollary 8 (Degree cancellation at $\alpha = 1$)**.** *If $\alpha = 1$, then*

$$x^\star_{ij} = H^{(1)}_{g_i g_j}. \tag{42}$$

The score is block-constant and contains no node-specific propensity. Every minimal factorization has center and context rows that depend only on community.

The $\alpha = 1$ result agrees with the NetMF target [Qiu et al., 2018]

$$\log\left[\mathrm{vol}\left(\sum_r a_r T^r\right)\mathcal{D}^{-1}\right] - \log k. \tag{43}$$

Right multiplication by $\mathcal{D}^{-1}$ already divides each conditional context probability by its stationary marginal. A second subtraction of both stationary marginals would count this normalization twice.

Corollary 9 (Additive context-degree term for $\alpha \neq 1$)**.** *For ordered SGNS and $\alpha \neq 1$, center-side rows of the score matrix remain block-constant. Context columns receive the additive term $(1-\alpha)\log\theta_j$. Moreover,*

$$X^\star = ZH^{(\alpha)}Z^\top + (1-\alpha)\mathbf{1}(\log\theta)^\top \tag{44}$$

has column space contained in $\mathrm{col}(Z)$ and rank at most K.

Proof. Equation (44) follows from Theorem 7. Because $\mathbf{1} = Z\mathbf{1}_K$,

$$X^\star = Z\left[H^{(\alpha)}Z^\top + (1-\alpha)\mathbf{1}_K(\log\theta)^\top\right]. \tag{45}$$

The column-space and rank claims follow. □

Some implementations tie center and context scores or combine both orientations of each pair. For a shared symmetric score, the pairwise optimum is

$$x_{ij}^{\mathrm{sym}} = \log \frac{2J_{ij}}{k(\pi_i q_j + \pi_j q_i)}. \tag{46}$$

Under the DCSBM,

$$x_{ij}^{\mathrm{sym}} = C_{ab} - \log \left(\theta_j^{\alpha-1} \kappa_a \kappa_b^{\alpha} \theta_i^{\alpha-1} \kappa_b \kappa_a^{\alpha} \right), \tag{47}$$

where C_{ab} is block-specific. The score is block-constant when $\alpha = 1$. For $\alpha \neq 1$, it is generally not multiplicatively separable; constant within-block propensities are a degenerate exception.

Remark 10 (Zeros and finite counts)**.** *If $F_{ab} = 0$, the unconstrained population score is $-\infty$. Finite-data methods must specify how they treat unobserved pairs, for example through truncation, smoothing, or a positive shifted-PMI convention.*

The pairwise optimum identifies dot products rather than the norm of either factor. For ordered SGNS, the optimal score has slope $1 - \alpha$ against log context degree within each context block. A law such as $\|u_i\| \propto d_i^{-\beta}$ requires an additional argument about low-rank approximation, optimization, and factor balancing.

7 Factorization gauge

The unconstrained SGNS loss depends on U, V only through $UV^\top$. For every invertible matrix R,

$$UV^\top = (UR)(VR^{-\top})^\top. \tag{48}$$

Individual norms and angles of U are therefore not identified by the objective. The invariance group contains all invertible transformations rather than only rotations.

Two restrictions make norm and angle claims well-defined. One can tie the factors, $U = V$, when the target admits the required positive-semidefinite factorization. Alternatively, one can impose the balance condition

$$U^\top U = V^\top V. \tag{49}$$

For a rank-h score matrix $M = P\Sigma Q^\top$, canonical balanced factors are

$$U_{\mathrm{bal}} = P\Sigma^{1/2}, \qquad V_{\mathrm{bal}} = Q\Sigma^{1/2}. \tag{50}$$

Equivalent minimal balanced factors differ only by an orthogonal transformation, which preserves row norms and angles. Empirical norm-degree claims should use balanced factors or identify the optimizer-dependent gauge under study.

8 Finite-sample transfer

The population theorems describe exact rays. Any row-wise embedding error bound yields corresponding bounds for radius and direction.

Lemma 11 (Stability of radius and direction)**.** *Suppose that population rows satisfy $u_i = r_i c_{g_i}$, and an estimated embedding obeys*

$$\|\widehat{u}_i Q - u_i\| \le \varepsilon_i \tag{51}$$

for an orthogonal alignment Q. Let $\rho_i = \|u_i\| = r_i \|c_{g_i}\| > 0$, and assume $\varepsilon_i < \rho_i$. Then

$$\left| \|\widehat{u}_i\| - \rho_i \right| \le \varepsilon_i \tag{52}$$

and

$$\left\| \frac{\widehat{u}_i Q}{\|\widehat{u}_i\|} - \frac{c_{g_i}}{\|c_{g_i}\|} \right\| \le \frac{2\varepsilon_i}{\rho_i}. \tag{53}$$

For two nodes,

$$\left| \cos(\widehat{u}_i, \widehat{u}_j) - \cos(c_{g_i}, c_{g_j}) \right| \le \frac{2\varepsilon_i}{\rho_i} + \frac{2\varepsilon_j}{\rho_j}. \tag{54}$$

Proof. The reverse triangle inequality gives Equation (52). For nonzero vectors x, y,

$$\left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\| \le \frac{2\,\|x - y\|}{\|y\|}. \tag{55}$$

Apply this inequality with $x = \widehat{u}_i Q$ and $y = u_i$ to obtain Equation (53). Equation (54) follows by adding the two unit-vector errors. □

Let $\Delta > 0$ be the minimum Euclidean distance between distinct population directions. Nearest-direction assignment is exact whenever

$$\max_i \frac{2\varepsilon_i}{\rho_i} < \frac{\Delta}{2}. \tag{56}$$

For the normalized-adjacency embedding, $\rho_i \propto \sqrt{\theta_i}$. The same absolute row error therefore produces more directional error for low-propensity nodes.

9 Relation to prior theory

Qin and Rohe [2013] derive the population ray structure of regularized spectral clustering under the DCSBM. Theorem 3 gives the unregularized structure in notation that also supports the walk calculations. Levy and Goldberg [2014] derive the shifted-PMI interpretation of SGNS, and Qiu et al. [2018] identify the matrix targets for DeepWalk, LINE, PTE, and node2vec.

Kojaku et al. [2024] connect a large-window linearization of the $p = q = 1$, $\alpha = 1$ score to the normalized Laplacian. Their linearized matrix has the form

$$\widehat{R} = D^{-1/2} \Gamma \Phi \Gamma^\top D^{-1/2}, \tag{57}$$

where Γ contains normalized-Laplacian eigenvectors and Φ transforms their eigenvalues. Equation (57) is a congruence rather than an ordinary eigendecomposition. The columns $U = D^{1/2}\Gamma$ solve

$$D\widehat{R}U = U\Phi, \tag{58}$$

which is a generalized eigenproblem. For a regular graph, D is proportional to the identity and the distinction disappears. It may also become negligible in a planted-partition regime where degrees concentrate around one value. Degree heterogeneity makes the distinction material.

Zhang and Tang [2022] prove exact community recovery for matrix-factorization versions of DeepWalk and node2vec under stochastic block models and degree-corrected variants. Davison et al. [2024] establish weakly consistent recovery for embeddings learned by node2vec and analyze the DCSBM case at unigram exponent one. These results establish community recovery under specific regimes. Our contribution concerns the radial-angular decomposition of the relevant population operators, the SGNS boundary, and the gauge required to interpret learned norms and angles.

10 Empirical predictions

The theory separates operators that standard node2vec training combines. A direct experiment should:

1. generate DCSBM graphs while retaining θ_i, g_i, and δ_i;
2. compute normalized-adjacency eigenvectors, a balanced singular-value factorization of raw stationary pair probabilities, a balanced factorization of the NetMF target, and trained SGNS center and context factors;
3. regress log norm on log expected degree within each true community;
4. test slopes 1/2, 1, and 0 for the first three population targets;
5. vary α explicitly and report center, context, and balanced SGNS norms separately; and
6. apply random invertible gauges to U, V while preserving $UV^\top$.

The population unit tests are exact. A balanced factorization of the raw pair matrix should have within-block slope one up to numerical tolerance. The $\alpha = 1$ score should have no within-block degree variation. Finite sampled graphs and trained embeddings require uncertainty estimates because walk sampling, diagonal removal, finite rank, and optimization introduce additional error.

Angular tests should measure within-community dispersion and between-community separation after row normalization. The stability bound in Lemma 11 predicts that low-degree nodes will have larger directional errors at the same absolute embedding error.

11 Scope and open problems

The population results support four claims. The normalized-adjacency eigenspace has $u_i = \sqrt{\theta_i/s_{g_i}}\, v_{g_i}$. The stationary raw pair matrix has $J_{ij} = \theta_i\theta_j G_{g_i g_j}$. Ordered SGNS has $x^\star_{ij} = H^{(\alpha)}_{g_i g_j} + (1-\alpha)\log\theta_j$. Unconstrained SGNS norms and angles require tying or balancing before they have an objective-level interpretation.

Several extensions remain open. General second-order node2vec requires formulas for the directed-edge state chain. A universal relationship between trained node2vec norm and observed degree would require control of finite-rank weighted approximation and optimization. Finite-sample community recovery from the radial-angular operators requires row-wise concentration bounds matched to the chosen operator. The loopless DCSBM also requires tracking the diagonal perturbation when expected degrees do not grow.

12 Discussion

The normalized-adjacency and raw-pair operators preserve degree through different radial exponents. The former produces a square-root relationship; the latter produces a linear relationship within each block. Their normalized directions remain community-specific.

SGNS changes the object being factorized. Marginal normalization at $\alpha = 1$ removes individual propensity from the population score. Other exponents introduce an asymmetric additive term on the context side. The unconstrained factorization then permits invertible changes of basis, so the objective alone does not determine row norms or angles.

These distinctions suggest a narrow empirical program. Each operator should be tested separately before trained SGNS embeddings are compared with the population geometry. Explicit control of the negative-sampling exponent and balanced-factor diagnostics can distinguish structural effects from an optimizer-dependent gauge.

13 Conclusion

Under the DCSBM population model, stationary random-walk joint probability has the form

$$J_{ij} = \theta_i \theta_j G_{g_i g_j}. \tag{59}$$

Every minimal exact factorization maps node i to $u_i = \theta_i c_{g_i}$, so expected degree controls radius within a community while normalized direction depends only on community. SGNS does not generally preserve this radial structure. At unigram exponent one, its marginal normalization removes node-specific degree; at other exponents, it introduces additive degree terms whose geometric effect depends on factorization and balancing.

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